ANSWER SHEET 2024-25 FALL

EE3014/3042 Midterm Exam | Date: 20 Nov, 2024

Student Name and Number

The notes and books will be kept closed. The calculator can be used. The necessary formulas are given on the last page.

1. (25p) A linear machine has magnetic flux density of 0.5 T directed into the page, a resistance of 0.25 Ω, a bar length of l = 1.0 m, and a battery of ~~100~~ 120 V.
   (a) If this bar has a load of 20 N attached to it opposite to the direction of motion, what is the steady-state speed of the bar?
   (b) If the bar runs off into a region where the flux density falls to 0.45 T, what happens to the bar? What is its final steady-state speed?
   (c) Suppose V_b is now decreased to 100 V with everything else remaining as in part (b). What is the new steady-state speed of the bar?
   (d) From the results for parts (b) and (c), what are two methods of controlling the speed of a linear machine (or a real dc motor)?
   (e) Assume that the motor is running at no-load conditions with a flux density of 0.5 T. What is the speed of the bar? Now apply a 30 N load to the bar. What is the new speed of the bar? What flux density would be required to restore the loaded bar to the same speed that it had under no-load conditions?

2. (25p) Suppose an eight-pole, 25-kW, 120-V dc generator has a duplex wave-wound armature which has 64 coils with 12 turns per coil. Its rated speed is 1800 r/min.
   (a) How much flux per pole is required to produce the rated voltage in this generator at no-load conditions?
   (b) What is the current per path in the armature of this generator at the rated load?
   (c) What is the induced torque in this machine at the rated load?
   (d) How many brushes must this motor have?
   (e) If the resistance of this winding is 0.012 Ω per turn, what is the armature resistance of machine?

3. (25p) A shunt dc generator in Figure 1 is rated at 6 kW, 120 V, 50 A, and 1800 r/min. Its magnetization curve is shown in Figure 2. This machine has the following characteristics: $R_A = 0.18\ \Omega, R_F = 20\ \Omega, V_F = 120\ V, N_F = 1000$ turns per pole, $R_{adj} = 0$ to $40\ \Omega$. If the variable resistor R_{adj} in this generator's field circuit is adjusted to $10\ \Omega$ and the generator's prime mover is driving it at 1800 r/min.
   (a) What is this generator's no-load terminal voltage?
   (b) Assuming no armature reaction, what is the terminal voltage of the generator with an armature cunent of 20 A?
   (c) Assuming an armature reaction equal to 300 A · turns at full load, what is the terminal voltage of the generator with an amature current of 20 A?
   (d) If the field resistor decreases to 5 Ω while the armature current remains 20 A, what will the new terminal voltage be? (Assume no armature reaction.)

4. (25p) In the speed control of a separately excited DC motor using armature voltage control, as shown in Figure 3, explain how the motor operates in two distinct regions: the constant torque drive region and the constant power drive region. Discuss the changes in torque (T_e), power (P), armature voltage (V_t), armature current (i_A), field current (i_f), and field flux (ϕ) across these two regions, and explain why each quantity behaves differently in each region.

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1. $B = 0{,}5\,T$ into page, $R = 0{,}25\,\Omega$, $L = 1\,m$, $V_B = 120\,V$

(25p) a) $F_L = 20N$, $V_{ss} = ?$ b) $B_1 = 0{,}45T$, $V_{ss,f} = ?$

c) $V_B = 100$ (decreased), $V_{ss,new} ?$

d) two methods of controlling the speed. ?

e) LDCM → at no load with $B = 0{,}5T$, $V_{ss} = ?$

Now $F_L = 30N$ to the bar, $V_s = ?$

$B = ?$ to bring the speed to the no load speed.

SOLUTION

5/ a) $F_{app} = F_{ind} = iLB \rightarrow i = \frac{F_{app}}{BL} = \frac{20N}{(0{,}5T)(1m)} = 40A$

$e_{ind} = V_B - iR = 120 - (40A)(0{,}25) = 110V$

$v_{ss} = \frac{e_{ind}}{BL} = \frac{110V}{(0{,}5T)(1m)} = 220\ m/s.$

5/ b) $T = 0{,}45T$, $F_{app} = F_{ind} = F_L = 20N \Rightarrow i = \frac{F_L}{BL} = \frac{20N}{(0{,}45T)(1m)}$

$\Rightarrow i = 44.44A \Rightarrow e_{ind} = V_B - iR = 120V - (44.44A)(0{,}25\Omega)$

$\Rightarrow e_{ind} = 108.88 \simeq 109\,V$

$V_{ss,f} = \frac{e_{ind}}{BL} = \frac{109V}{(0{,}45T)(1m)} \simeq 242\ m/s.$

5/ c) $V_B = 100V \rightarrow F_L = F_{ind} = 20N$ (finally)

$F_{ind} = F_L = iLB \Rightarrow i = \frac{F_L}{BL} = \frac{20N}{(0{,}45T)(1m)} = 44.44A$

$e_{ind} = V_B - iR = 100V - (44.44A)(0{,}25\Omega) = 88{,}9\,V$

$v_{ss} = \frac{e_{ind}}{BL} = \frac{88{,}9V}{(0{,}45T)(1m)} \simeq 197.5\ m/s.$

↑ continued) d) 5/ Two methods: → Changing flux density B
→ changing Battery voltage V_B

e) (5) $B = 0{,}5T$. $e_{ind} = V_B$ at no load.

$$v_{nl} = \frac{e_{ind}}{Bl} = \frac{100V}{(0{,}5T)(1m)} = \underline{200\ m/s}.$$

$$\rightarrow F_L = F_{ind} = 30N \rightarrow i = \frac{F_{ind}}{Bl} = \frac{30N}{(0{,}5T)(1m)} = \underline{60A}.$$

$$e_{ind} = V_B - iR = 100V - (60A)(0{,}25\Omega) = \underline{85V}$$

$$v_{ss} = \frac{e_{ind}}{Bl} = \frac{85V}{(0{,}5T)(1m)} = \underline{170\ m/s}.$$

V_{nl} ↓ V_{ss} 3/

$$200 = \frac{e_{ind}}{Bl} = \frac{V_B - iR}{Bl} = \frac{V_B - (F_L/Bl)R}{Bl} = \frac{100 - \frac{30}{B}(0{,}25)}{B}$$

$$200B^2 - 100B + 30(0{,}25) = 0 \rightarrow B^2 - 0{,}5B + 0{,}0375 = 0$$

$$\rightarrow \underline{B_1 = 0.4081\ T},\ \underline{B_2 = 0{,}0919\ T} \rightarrow \text{not possible}$$ (because the numerator becomes negative)

2) (25p) $P = 8$, $P_{out} = 25\ kW$, 120V DC generator, duplex wave wound armature winding ($m = 2$)

$C = 64$, $N_c = 12$, $n_{rated} = 1800\ r/min$.

a) ϕ/pole ? to produce V_{rated}, b) I_A/path at R_{FL} or at rated load.

c) τ_{ind} at rated load ? d) # of brushes ? e) $R_W = 0{,}012\Omega$ per turn, R_A = ?

SOLUTION $V_T - I_A R_A = E_A \rightarrow R_A$ is ignored since it is not given.

a) $E_A = K\phi\omega = \frac{ZP}{2\pi a}\phi\omega$, $a = 2 \times 2 = 4$ number of parallel paths (duplex, wave winding)

7/10 Z ← # of conductor $= (C\ coils)(12\ turns/coil)(2\ conductors/turn) = (64)(12)(2) = \underline{1536}$

$$E_A = \frac{ZP}{2\pi a}\phi\omega = \frac{(1536)(8)}{(2\pi)(4)}\phi\left(1800\frac{2\pi}{60}\right) = 120V$$

(across armature winding; $1800\frac{2\pi}{60} = 30(2\pi)$)

$\rightarrow \phi = 1.3 \times 10^{-3}$ Weber/pole

3/6

2 continued)

b) 6/ $I_A = \frac{P_{out}}{V_{rated}} = \frac{25kW}{120V} = 208,3A$

since $a = 4$, Hence → $I = \frac{I_A}{a} = \frac{208,3}{4} \cong 52A$ / path

c) 5/ $\tau_{ind,rated} = \frac{\frac{Z_p}{2\pi a}}{K} \Phi I_A = \frac{(1536 \text{ conductors})(8 \text{ poles})}{2\pi (4 \text{ paths})} (1,3 \times 10^{-3}) (208,3)$

$\tau_{ind,rated} = 132,4 Nm$

2nd method (approximately)

$\tau_{ind} = \frac{P_{out}}{\omega} = \frac{25kW}{1800 \frac{2\pi}{60}} = 132,6 Nm$

Parts d and e is on page 6.

3.) DC Shunt GEN, $P_{out} = 6kW$, 120V, 50A, 1800 r/min

(25p) $R_A = 0,18\Omega$, $R_F = 20\Omega$, $V_F = 120V$, $N_F = 1000$ turns/pole

$R_{adj} = 0$ to 40Ω $R_{adj} = 10\Omega$ (set)

a) $V_{T,no\,load} = ?$ b) $\mathcal{F}_{AR} = 0$, $V_T = ?$ at $I_A = 20A$

c) $\mathcal{F}_{AR} = 300$ A turns, at full load, $V_T = ?$ at $I_A = 20A$

d) Assume $R_f = 5\Omega$. $I_A = 20A$, $\mathcal{F}_{AR} = 0$, $V_T = ?$

SOLUTION a) $R_f + R_{adj} = 30\Omega \rightarrow I_F = \frac{V_F}{R_t} = \frac{120}{30} = 4A$

5/ $V_T \cong 121V$ (no load) $= E_A$ (from the magnetization curve)

5/ b) $V_T = V_{T,no\,load} - (R_A)(I_A) = 121V - (0,18)(20) = 117.4V$

c) $\mathcal{F}_{AR} = 300$ A turns, $I_A = 20A$, $V_T = ?$ 9/

$I_{eff} = I_F - \frac{\mathcal{F}_{AR}}{N_F} = 4A - \frac{300 \text{ A turns}}{1000 \text{ turns}}$

$I_{eff} = 4 - 0,3 = 3.7A$

From the Magnetization curve $E_A \cong 118V$

At $I_A = 20A \Rightarrow V_T = E_A - \underbrace{R_A I_A}_{3.6V} = 118V - 3,6V = 114.4V$

d) $R_{adj} = 5\Omega \rightarrow I_F = \frac{V_F}{R_f + R_{adj}} = \frac{120}{25} = 4,8A$, from MC → $E_A \cong 128V$ 6/

$V_T = E_A - R_A I_A = 128V - (0,18\Omega)(20A) = 124.4V$

You can only change field resistance by R_{adj}

Separately excited DC motor

(4) 4) A. The constant torque drive region ($W < W_{base}$)

(25p) 25/ a) The motors operates with a constant torque (T_e). $T_e \propto i_{a_t}$, $T_e \propto \phi$. Since the field flux is kept constant, T_e is directly controlled by i_a. The load is held constant.

25/ b) The power (P) increase linearly with increasing the speed.

$$P = T_e W \quad \text{where } T_e \text{ is constant}$$

25/ c) The speed is controlled by varying the armature voltage(V_t)

$\uparrow V_t = \uparrow E_A + R_A i_a$ hence $\uparrow E_A = K\phi W \uparrow$ So, $\uparrow V_t \uparrow W$.

25/ d) Armature currents(i_a) remains constant because the torque demand does not change, the torque is directly proportional to i_a.

25/ e) Field currents(i_f) and field flux(ϕ) both remain constant, a field weakining is not active in this region.

B. The Constant power drive region ($W > W_{base}$)

a) Torque(T_e) decreases as the speed increases beyond base speed.

25/ Since Power(Since V is at rated value) is constant in this region, $T_e = \frac{P}{W}$, inversely proportional to the speed(W)

25/ b) Armature voltage(V_t) remains constant at its rated value. Further increase in speed is achived by reducing the flux.

25/ c) Armature current (i_a) remains constant ($P = V_t i_a$ constant) while the torque decreases as the flux decrases that speeds up the shaft.

25/ d) Field current (i_f) and Flux (ϕ): In this region, field weakining is activated by reducing i_f. This causes to reduce the ϕ, allowing the motor to achive higher speed beyond the base speed. V_t is constant in this region.

2.5/ e) The power (P) is constant. The motor operates at a constant power. As speed increases, torque decreases while the power remains constants. 5/6

2 continued

d) The number of parallel paths(a) is always 2, not up to the number of poles.

3/ For a duplex winding, the number of parallel paths is equal to $2a = (2)(2) = 4$ (doubles) = # of brushes

e) $N = 12$, $C = 64$,

4/ Total turns $= 64 \times 12 = 768$

of turns per path, $N_p = \frac{768}{16} = 48$ turns/path

The total resistance per path $R_p = (48)(0,011) = 0,528\,\Omega$

The armature resistance, $R_A = \frac{R_p}{a} = \frac{0,528}{4} = 0,132\,\Omega$

